FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – VIII
PAPER –2
TEST DATE: 05-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.
  Section – A *(01 – 04, 18 – 21, 35 – 38)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
  Section – A *(05 –07, 22 – 24, 39 – 41)*: This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
  Section – B *(08 – 13, 25 – 30, 42 – 47)*: This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.
  Section – C *(14 –17, 31 – 34, 48 – 51)*: This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Physics

PART – I

SECTION – A
(One Options Correct Type)

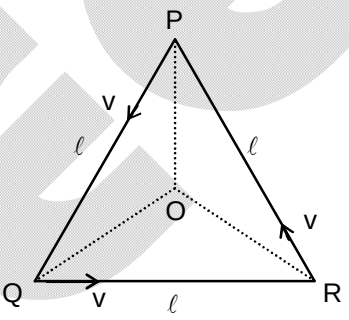
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

1. The position vector of a point P moving in space is given by $\vec{r} = 3\cos t(\hat{i}) + 4\cos t(\hat{j}) + 5\sin t(\hat{k})$. Find the time when the point P crosses the plane 4x – 3y + 2z = 5.

(A) $\frac{\pi}{2}$ sec (B) $\frac{\pi}{3}$ sec

(C) $\frac{\pi}{4}$ sec (D) $\frac{\pi}{6}$ sec

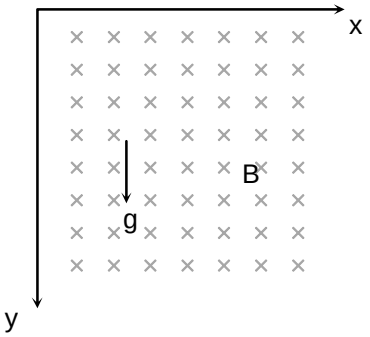
2. Three particles P, Q and R initially located at the vertices of an equilateral triangle of side ℓ start moving with constant speed v each such that P always chases Q, Q always chases R and R always chases P. If 'O' is the centroid of the triangle then find the distance covered by any particle in one revolution around centroid 'O' of the triangle.



(A) $\frac{\ell}{3}\left(1-e^{-\sqrt{3}\pi}\right)$ (B) $\frac{\ell}{3}\left(1-e^{-2\sqrt{3}\pi}\right)$

(C) $\frac{2\ell}{3}\left(1-e^{-\sqrt{3}\pi}\right)$ (D) $\frac{2\ell}{3}\left(1-e^{-2\sqrt{3}\pi}\right)$

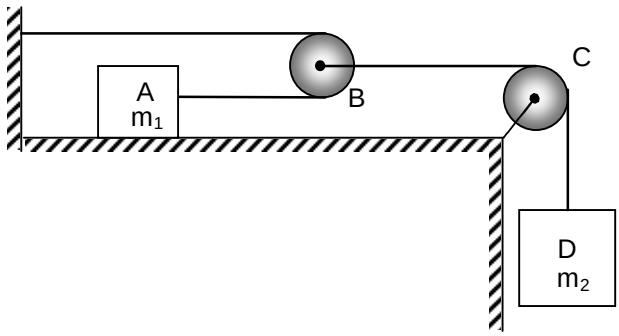
3. A constant horizontal uniform magnetic field B = 5×10^6 T exist in a uniform vertical gravitational field g = 10 m/s^2. Both fields are perpendicular to each other. A particle of mass (m = 50gm) and charge (q = 2μC) is released in the space. Find the displacement of particle along y-axis when velocity of particle is along x-axis for the first time.



(A) 5 cm (B) 10 cm

(C) 15 cm (D) 20 cm

4. In the arrangement shown in the figure m_1 = 15 kg, m_2 = 20 kg and mass per unit length of the string AB is 30×10^{-3} kg/m. At t = 0, the system is released form rest and at the same instant of time a transverse pulse is produced at A. The pulley B is at a distance ℓ = 10 m from A initially. Find the time taken by the pulse to reach the pulley B. (Take g = 10 m/s^2 and $\sqrt{102} = 10.2$)



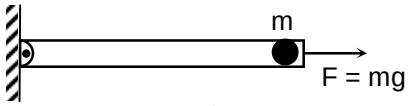
(A) 0.1 sec (B) 0.2 sec

(C) 0.4 sec (D) 0.8 sec

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A thin uniform rod of mass 'm' and length 'L' is hinged at one end and is free to rotate in the vertical plane A small ball of mass m is fixed to other end of the rod. A constant horizontal force F = mg is also acting on the rod at the free end when the rod is released from horizontal position as shown in the figure. Choose the correct option(s).



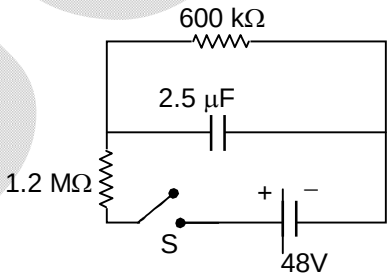
(A) Accelertion of the ball at the instant of releasing the rod is $\frac{9g}{8}$.

(B) The sum of horizotnal and vertical component of reaction froce exerted by the hinge when the rod has just been released is $\frac{5mg}{16}$

(C) The sum of horizontal and vertical component of reaction force exerted by the hinge when the rod has just been released is $\frac{21}{16}$mg

(D) The speed of the ball when the rod becomes vertical is $\sqrt{\frac{3g\ell}{4}}$

6. In the circuit shown, capacitor is initially uncharged and switch S is closed at t =0. Choose the correct option(s).



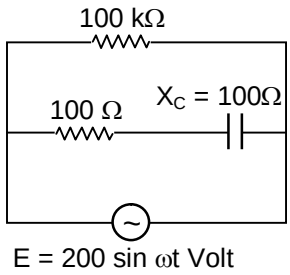
(A) Initially current through battery just after 'S' is closed is 40 μA

(B) The time after which current through capacitor becomes half of its initial value is 4 sec.

(C) The current through the 600 kΩ resistor as a function of time is $\frac{80}{3}\left(1-e^{-t}\right)\mu A$

(D) The current through capacitor long time after 'S' is closed is zero.

7. In the RC circuit shown below. Choose the correct option(s).



E = 200 sin ωt Volt

(A) The power factor of the circuit is $\frac{3}{\sqrt{10}}$

(B) The average power dissipated is 300 W

(C) The power factor of the circuit is $\frac{1}{\sqrt{10}}$.

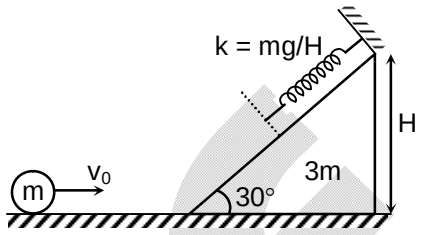
(D) The phase difference between the current through the source and source voltage is $\frac{\pi}{4}$

SECTION – B

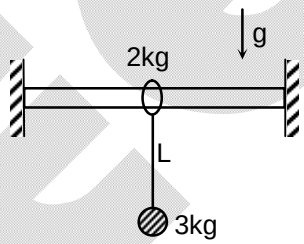
(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

8. A block of mass 'm' is pushed towards a movable wedge of mass '3m' and height H with a highly elastic spring extended upto its half of its slant length. If the minimum value of v_0 required to compress the spring completely is $\sqrt{NgH}$. (Here $k = \frac{mg}{H}$). Find the value of N.

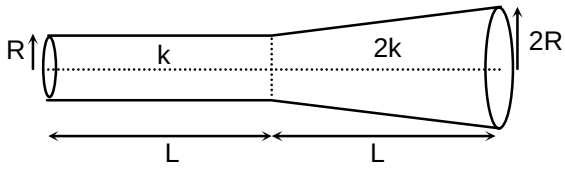


9. A small ring of mass 2 kg can move smooth on a fixed horizontal rod. This ring is connected to point mass 3 kg with the help of a massless string of length 'L' as shown in the figure. If the time period for small oscillations of this system is $T = 2\pi\sqrt{\frac{mL}{5g}}$, then find the value of 'm'.

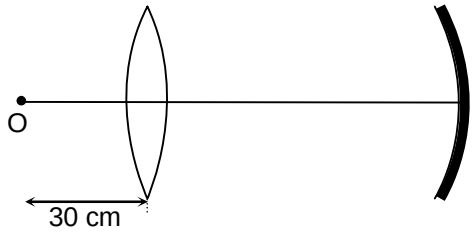


10. The diameter of a wire of length 100 cm is measured with the help of a screw gauge. The main scale reading is 1 mm and circular reading is 25. Pitch of the screw gauge is 1 mm and total member of divisions on the circular scale is 100. The wire is used in a experiment for determination of Young's modulus of the wire by Searl's method. The following data is available, elongation in the wire $\Delta\ell$ = 0.125 cm under the tension of 50 N, least count for measuring normal length of wire is 0.01 cm and for elongation in the wire is 0.001 cm. Assuming that the force is measured accurately, the maximum percentage error in calculating the value of Young's modulus (Y) is $\frac{8\times n}{10}$%, where n is very nearly an integer, find n.

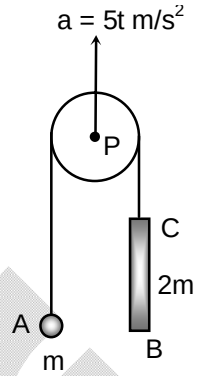
11. A composite object is formed by combining a uniform rod of circular cross section with thermal conductivity k and a frustum of same length with thermal conductivity 2k as shown in the figure. If equivalent thermal conductivity of the composite body is $\frac{6k}{\alpha}$, find α.



12. A lens of focal length 20 cm is placed co-axially with a concave mirror of radius of curvature 20 cm. A point object 'O' is placed at a distance of 30 cm from the lens on its principal axis as shown. The final image formed after two refractions from lens one reflection by mirror is coincident with the object 'O' itself. If α be the maximum distance of mirror from object 'O' and β be the minimum distance of mirror from the object, then find the value of $\alpha + \beta$ (in cm).



13. There is a massless pulley which is going upward with an acceleration a = 5t m/s^2 where t is time in seconds. A string is passing over the pulley as shown in the figure. At one end of the string a plank of mass 2m kg is attached and at the other end a small ball of mass m kg is attached. It is given that after 1 sec small ball just passes other end of the plank. If length of the plank is ℓ (in meter), find 9ℓ. (Take g = 10 m/s^2)



SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 14 and 15

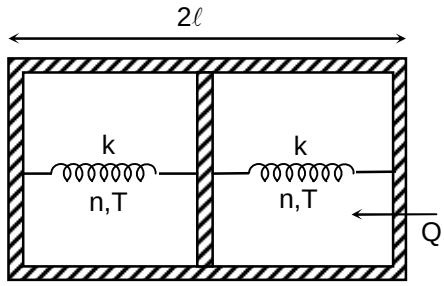
A radioactive nuclei X at rest disintigrates into two nucleus Y and Z mass of Y is 12 m and that of Z is 4m. The Q-value of the reaction is $Q = \dfrac{h^2}{24m\lambda^2}$, where h is Plank's constant and λ is constant. The energy liberated in reaction is completely imparted to products Y and Z as kinetic energy.

14. If the mass of nucleus x is $\alpha m + \dfrac{h^2}{\beta m C^2 \lambda^2}$, then find the value of $\beta - \alpha$.

15. If de-Broglie wavelength of Y is $n\lambda$, then find the value of n.

Paragraph for Question Nos. 16 and 17

A horizontal isolated cylindrical vessel of length 2ℓ is separated by a thin insulating piston into two equal parts each of which contains n moles of an ideal monoatomic gas at temperature T. The piston is connected to the end faces of the vessel by un deformed spring of force constant k each. The left part is in contact with a thermostat (a device which maintains initial and final temperature constant). When an amount of heat Q is supplied to the gas in right part, the piston is displaced to the left by a distance $\dfrac{\ell}{2}$.



16. If $\dfrac{k\ell^2}{2} = 500$ J, T = 100 K, n = 3 and $R = \dfrac{25}{3}$ J/mol-k is given then find the change in temperature of right part of the cylindrical vessel in K is ……

17. If Q = 12000 J, $\dfrac{k\ell^2}{2} = 500$ J, n = 3, $R = \dfrac{25}{3}$ J/mol-k and T = 100 K then the heat given away at temperature T to the thermostat by the left past of the piston in J is……..

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

18. Which of the following species has/have same number of X – O – X linkages where X = S or P or B.

(I) $\gamma - SO_3$

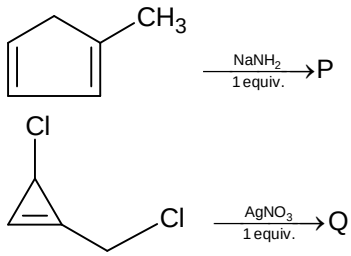
(II) P_4O_6

(III) $Na_2B_4O_7.10H_2O$

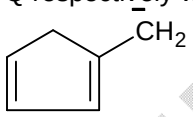
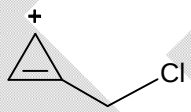
(IV) $(HPO_3)_3$

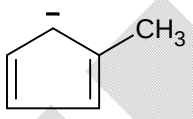
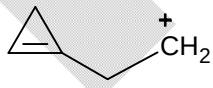
(A) II and III (B) I and III

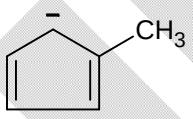
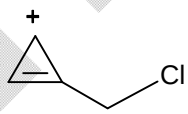
(C) I and IV (D) II and IV

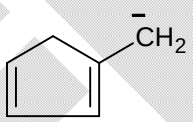
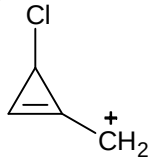
19. 

P and Q respectively will be

(A)  and 

(B)  and 

(C)  and 

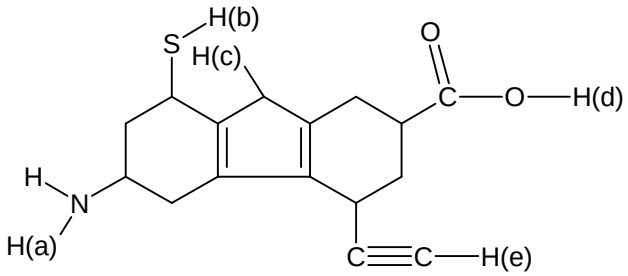
(D)  and 

20. If subsidiary quantum number of a sub energy level is 5, the maximum and minimum values of the spin multiplicities are

(A) 10, 1 (B) 12, 1

(C) 10, 2 (D) 8, 1

21.



Arrange the marked hydrogens according to increasing acidic strength.

(A) e < a < c < b < d (B) e < a < c < d < b

(C) a < e < c < b < d (D) a < e < c < d < b

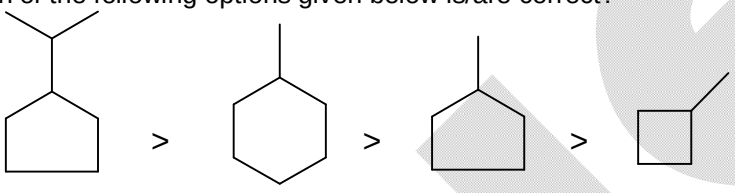
SECTION – A

(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

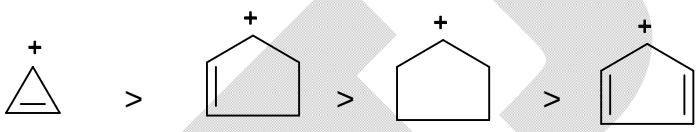
22. Which of the following options given below is/are correct?

(A)



(Order of heat of combustion)

(B)



(Order of stability of carbocation)

(C)



(Order of heat of hydrogenation)

(D) $\dot{C}H_3 < \dot{C}H_2F < \dot{C}HF_2 < \dot{C}F_3$

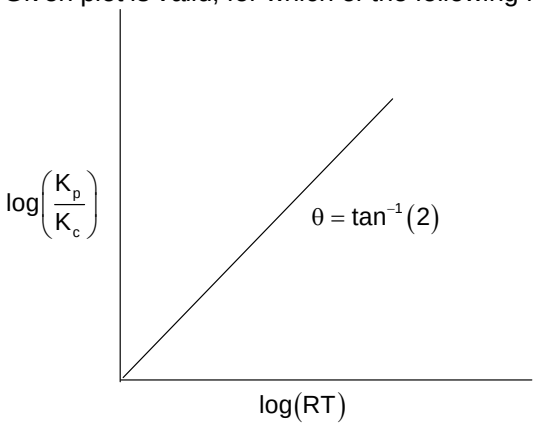
(Increasing s-character of the orbital containing lone electron)

23. Fac-mer isomerism is possible in

(A) $\left[M(NH_3)_3(H_2O)_3\right]$ (B) $\left[M(gly)_3\right]$

(C) $\left[M(dien)Cl_3\right]$ (D) $\left[M(en)_3\right]$

24. Given plot is valid, for which of the following reaction



(A) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

(B) $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$

(C) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(D) $CuSO_4.5H_2O(s) \rightleftharpoons CuSO_4.3H_2O(s) + 2H_2O(g)$

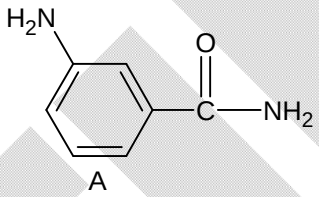
SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

25. The heat of combustion of $PH_3(g)$ and $P_2H_4(g)$ are 954 kJ mol^{-1} and 1485 kJ mol^{-1} respectively. The P – P bond energy in kJ mol^{-1} is _____.

26. Volume of 0.5 M H_2SO_4 (in ml) required to neutralize all NH_3 formed, when 2.72 g of the substance A is treated for Kjeldahl's method of N-estimation is_____



A

27. 10 L polluted air at a temperature of 27°C and 0.82 atm pressure was passed through lime water giving 10 g $CaCO_3$. Value of pH of lake water that is located on the polluted site is_____

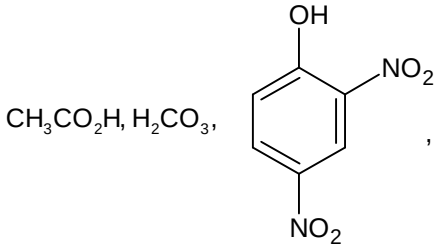
$K_{a_1}(H_2CO_3) = 10^{-8}$

$K_{H_{CO_2}} = 3.333 \times 10^{-2}$ M atm^{-1}

28. If the slope of Z versus P curve is constant $\left(\text{slope} = \frac{\pi}{49.26} atm^{-1}\right)$ at temperature 300 K and at very high pressure, then diameter of the molecule is _____pm.

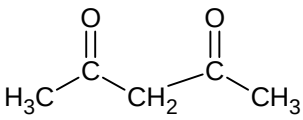
[Assume $N_A = 6 \times 10^{24}$, $R = 0.0821$ L atm mol^{-1} K^{-1}]

29. How many of the following compounds are stronger acid than benzoic acid?

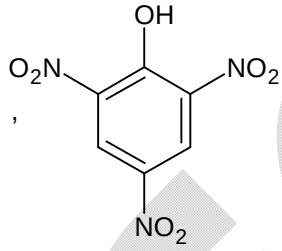


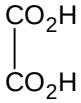
CH_3CO_2H, H_2CO_3, , $H_3C—CH_2—SH$,

$HO_2C-CH_2-CO_2H$,



,





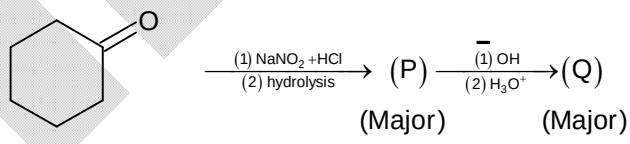
, $F_3C—CH_2—OH$

30. How many of the following statements are correct about nobel gases.
    1. The solubility of noble gases in water increases down the group.
    2. Boiling point increases down the group.
    3. Radon has no stable isotope.
    4. Helium cannot be solidified by cooling at atmospheric pressure.
    5. All of them form clathrate compounds with hydroquinone.
    6. Xenon fluorides are hypervalent.
    7. KrF_2 is the first chemical compound a noble gas.

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 31 and 32



31. Percentage of carbon in 'P' would be (approximately)

32. If molar mass of Q is 'x', then $\frac{x}{20}$ would be

Paragraph for Question Nos. 33 and 34

Oxidising power of $KMnO_4$ depends on pH of the medium. In neutral or faintly alkaline solution $KMnO_4$ is stronger oxidizing agent than in acidic medium.

$MnO_4^- + 8H^+ + 5e \longrightarrow Mn^{2+} + 4H_2O \qquad E^o = 1.52\ V$

$MnO_4^- + 4H^+ + 3e \longrightarrow MnO_2 + 2H_2O \qquad E^o = 11.69\ V$

33. Mols of I^- can be oxidized by using 1000 ml 0.5 M $KMnO_4$ in faintly alkaline medium would be

34. Mols of I^- can be oxidized by using 500 ml of 0.1 M $KMnO_4$ in acidic medium would be

FIITJEE

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. If 7 balls are drawn one by one with replacement from a bag containing 5 balls numbered 0, 2, 3, 4, and 5, then number of ways in which a sum of 30 can be obtained, is not a multiple of
(A) 3 (B) 5
(C) 7 (D) 8

36. If $\sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left[\frac{mn}{3^n}\left(\frac{1}{3^m}-\frac{n}{n\cdot 3^m+m\cdot 3^n}\right)\right]=\frac{p}{q}$ (where p and q are relatively prime), then q – 3p is
(A) 2 (B) 3
(C) 4 (D) 5

37. $\prod_{n=3}^{\infty}\left(1-\tan^4\frac{\pi}{2^n}\right)$ is equal to
(A) $\frac{\pi^3}{8}$ (B) $\frac{\pi^3}{16}$
(C) $\frac{\pi^3}{32}$ (D) $\frac{\pi^3}{64}$

38. If $f(x) = ax^{17} + b\sin x \sin 2x \sin 3x + cx^2\,\text{sgn}(\sin x) - d\log_e\left(\sqrt{1+x^2}-x\right) + x(|x+1| - |x-1|)\left(\frac{e^{2x}-1}{e^{2x}+1}\right)$ be defined on the set of real numbers, $(a > 0, b, c, d \in R)$, if $f(-7) = 7$, $f(-5) = -5$, $f(-2) = 3$, then the minimum number of zeroes of the equation $f(x) = 0$ is
(A) 6 (B) 7
(C) 8 (D) 9

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. If f(x), g(x) and h(x) are polynomials of degree 2 and $\Delta(x)=\begin{vmatrix} f(x) & g(x) & h(x) \\ f(x-a) & g(x-a) & h(x-a) \\ f(x-b) & g(x-b) & h(x-b)\end{vmatrix}$, then
(A) $\Delta'(x)$ is independent of x
(B) $\Delta''(x)$ is independent of x but dependent on a and b
(C) $\Delta''(x)$ is independent of x, a and b
(D) $\Delta'''(x)$ is independent of x, a and b

40. A cubic equation with real coefficients p(x) = 0 has one real root (α) and two complex roots ($\beta \pm i\gamma$) represented by A, B and C on the argand plane respectively. Then
(A) roots of p′(x) = 0 are complex if A lies inside one of the two equilateral triangles with base BC
(B) roots of p′(x) = 0 are real if A lies inside one of the two equilateral triangles with base BC
(C) roots of p′(x) = 0 are complex if A, B, C are vertices of an isosceles (not equilateral) triangle
(D) roots of p′(x) = 0 are real and equal if $\frac{\alpha-\beta}{\gamma}=\pm\sqrt{3}$

41. A non-zero vector $\vec{a}$ is given by $\vec{a}=\vec{b}\times\vec{c}+2\vec{b}$ where $|\vec{b}|=|\vec{c}|=2$ and $|\vec{a}|\le 4$, then sum of possible values of $|2\vec{a}+\vec{b}+\vec{c}|$ is divisible by
(A) 2 (B) 3
(C) 4 (D) 5

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. If 6 digit numbers are formed using the digits 1 and 8 only and n_1, n_2, n_3 are the numbers of such 6 digits numbers which are divisible by 231, 84 and 63 respectively, then $n_3 - n_2 - n$, is

43. If $\int\frac{(e^x-1)\sin x-(e^x-1-x)\cos x}{1+(e^x-1-x-\cos x)(e^x-1-x+\cos x)}=f[g(x)]+c$ where c is constant of integration and $\lim\limits_{x\to0}g(x)=0$. If $\lim\limits_{x\to0}\left(\frac{f(x)}{x}+\frac{g(x)}{x}\right)=\frac{m}{n}$ where m, n $\in$ N, then the minimum value of m + n is

44. Let ABCD be a square with vertices A(0, 0), B(4, 0), C(4, 4), D(0, 4). The square is folded in such a way that corner B always lies on AD. As B moves on AD, the crease thus formed always touches a fixed parabola whose length of latus rectum is

45. If the area bounded by the curves $y=x+\frac{2}{3}x^3+\frac{2}{3}\cdot\frac{4}{5}x^5+\frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}x^7+.....\infty$ (where |x| < 1) and x = x_1, x = x_2 and y = 0 such that x_1 and x_2 are the roots of $4x^2-2\left(2\tan\frac{5\pi}{12}-1\right)x+\sqrt{3}=0$ is given by $\frac{\pi^a}{b}$, where a, b $\in$ N, then value of (a + b – 19) is

46. If z_1 and z_2 are two different roots of the equation $z^{42}=1$ (where z is a complex number) and the probability that $|z_1+z_2|^2\ge\tan\frac{5\pi}{12}$ is P, then the value of 41P is

47. Let f : (0, ∞) → (0, ∞) be a twice differentiable function such that $f\left(\tan\frac{\pi}{4}\right)=1+\tan\frac{\pi}{8}$ and $f(x)\cdot f'\left(\frac{1}{x}\right)=1$ for all the positive value of x, then $[f(3)]^2 - 2$ is

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 48 and 49

The line $x + 2y + a = 0$ intersect the circle $x^2 + y^2 = 4$ at points A and B. Another line $12x – 6y – 41 = 0$ intersect the circle $x^2 + y^2 – 4x – 2y + 1= 0$ at two distinct points C and D. Let another circle $S = 0$ is passing through the points A, B, C and D

48. If the equation of circle $S = 0$ is $x^2+ y^2 + 2gx + 2fy + c = 0$, then $|2g + 2f + c|$ is

49. Values of a is

Paragraph for Question Nos. 50 and 51

Given $f(x + y + z) = f(x) f(y) f(z)$ $\forall$ $x, y, z \in R$ and $f(2) = 4$ and $f'(0) = 3$

50. Number of possible value(s) of f(0) is/are

51. Sum of digits of $f'(2)$ is